

# SQL Window Value Functions

17 · Window Value Functions — Complete Reference Guide

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## 1 & 2. LEAD & LAG — Row Comparison

→ *Task 1: Month-over-Month Sales change using LAG*

```
SELECT
    *,
    CurrentMonthSales - PreviousMonthSales AS MoM_Change,
    ROUND(
        CAST((CurrentMonthSales - PreviousMonthSales) AS FLOAT)
        / PreviousMonthSales * 100, 1
    ) AS MoM_Perc
FROM (
    SELECT
        MONTH(OrderDate) AS OrderMonth,
        SUM(Sales)        AS CurrentMonthSales,
        LAG(SUM(Sales)) OVER (ORDER BY MONTH(OrderDate)) AS PreviousMonthSales
    FROM Sales.Orders
    GROUP BY MONTH(OrderDate)
) AS MonthlySales;
```

→ *Task 2: Customer Loyalty — rank by average days between orders using LEAD*

```
SELECT
    CustomerID,
    AVG(DaysUntilNextOrder) AS AvgDays,
    RANK() OVER (
        ORDER BY COALESCE(AVG(DaysUntilNextOrder), 999999)
    ) AS RankAvg
FROM (
```

```

SELECT
    OrderID, CustomerID,
    OrderDate AS CurrentOrder,
    LEAD(OrderDate) OVER
        (PARTITION BY CustomerID ORDER BY OrderDate) AS NextOrder,
    DATEDIFF(
        day,
        OrderDate,
        LEAD(OrderDate) OVER
            (PARTITION BY CustomerID ORDER BY OrderDate)
        ) AS DaysUntilNextOrder
FROM Sales.Orders
) AS CustomerOrdersWithNext
GROUP BY CustomerID;

```

## 3 & 4. FIRST\_VALUE & LAST\_VALUE

→ *Task 3: Lowest and highest sales per product, and deviation from lowest*

```

SELECT
    OrderID, ProductID, Sales,
    FIRST_VALUE(Sales) OVER (
        PARTITION BY ProductID
        ORDER BY Sales
    ) AS LowestSales,
    LAST_VALUE(Sales) OVER (
        PARTITION BY ProductID
        ORDER BY Sales
        ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING
    ) AS HighestSales,
    Sales - FIRST_VALUE(Sales) OVER (
        PARTITION BY ProductID
        ORDER BY Sales
    ) AS SalesDifference
FROM Sales.Orders;

```